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B. E. (Sixth Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(Mechanical Engg. Branch)

INTERNAL COMBUSTION ENGINES

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all five questions. Part (a) is compulsory. Attempt any two parts from (b), (c) and (d). Explain answers with neat diagrams.*

1. (a) What is the difference between internal combutstion engine and external combustion engine? 2

(b) Explain the working principle of two stroke engine and four stroke engine with the help of neat diagrams. 7

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(c) Draw the actual value timing diagram for four stroke petrol engine. Explain the reasons why actual value timing diagram is deviated from theoletical value timing diagram. 7

(d) Compare S. I. engine and C. I. engine at the following points :

(i) Basic cycle

(ii) Fuel used

(iii) Induction of fuel

(iv) Load control

(v) Ignition

(vi) Compression ratio

(vii) Speed 7

2. (a) Define octane number. 2

(b) State the process of combustion in C. I. engine. Explain different stages of combustion in C. I. engine. 7

(c) What is ASTM distillation curve? Write effect of volatility on :

(i) Cold starting

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(ii) Carburettor icing

(iii) Crank case dilution

(d) List the important qualities of S.I. engine fuel and explain them. 7

3. (a) Define carburation.

(b) Derive the expression for air, fuel ratio from single carburettor, taking compressibility of air into account. 7

(c) Draw neat sketch of simple carburettor and explain its working. What are the limitations of simple carburettor? 7

(d) A simple jet carburettor has to supply 5 kg of air per minute. The air is at a pressure of 1·013 bar and at a temperature of 27 °C. Calculate the throate diameter of the choke for air flow velocity of 90 m/s. Take velocity coefficient to be 0·8. Consider isentropic flow. Assume air as compresable fluid Consider $R = 0{\cdot}287$, $C_p = 1{\cdot}005$ kJ/kg K, $r = 1{\cdot}4$ for air. 7

4. (a) What is meant by firing older? 2

(b) Discuss the advantages and limitations of air cooling system and water cooling system. 7

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(c) With the help of neat diagram explain the working principle of Bosch fuel in section pump. 7

(d) Explain dry sump lubrication system and wet sump lubrication system used is internal combustion engines. 7

5. (a) Define volumetric efficiency. 2

(b) What is meant by friction power? Explain Morse test and Willan's line method to measure friction power. 7

(c) The following details were noted is a test on a single-cylinder, four stroke engine :

bore = 150 mm, stroke= 160 mm, speed of the engine = 500 rpm, fuel consumption = 0·0475 kg per minute, calorific value of the fuel is 42,000 kJ/kg, the difference in tension on either side of the brake Pulley is 400 N. brake circumference is 2·2 m, length of the indicator diagram is 50 mm, area of the indicator diagram is 450 mm^2, spring constant is 0·8333 bar per mm.

Calculate :

(i) brake power

(ii) indicated power

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(iii) mechanical efficiency

(iv) brake thermal efficiency 7

(d) The following observations were made during the test on an oil engine.

Brake power is 31·5 kW, fuel consumed 10·5 kg/hr. Calorific value of the fuel is 43,000 kJ/kg, circulating water for cooling is 540 kg/hr, rise in the temperature of cooling water is 56 °C, water circulated through exhaust gas calorimeter is 454 kg/hr, Rise in the temperature of water passing through exhaust gas calorimeter is 36 °C. Temparature of exhaust gases leaving the exhaust gas calorimeter is 82 °C. Ambient air temperature is 17 °C Air fuel ratio 19 : 1 C_p for exhaust gases 1 kJ/kg K.

Prepare heat balance sheet on percentage basis. 7

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